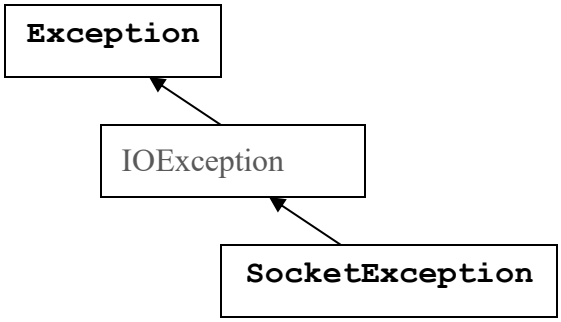
Question 1



The inheritance hierarchy diagram on the right shows the class relationships between various exception classes. For the **`try`** block below, if we wish to **`catch`** each type of exception object separately, in what order, top to bottom, may we place these **`catch`** blocks.

```
try
{
  Scanner keyboard = new Scanner(System.in);
  int count = keyboard.nextInt();
  if      (count == 1) throw new IOException();
  else if (count == 2) throw new Exception();
  else                 throw new SocketException();
}
```

Question 2

Circle the letters next to the corresponding statements below that are always ***false***.

a) A non-**`static`** method from **`ClassX`** may call another non-**`static`** method from **`ClassX`**.

b) A non-**`static`** method from **`ClassX`** may call a **`public static`** method from **`ClassZ`**.

c) A non-**`static`** method from **`ClassX`** may call a **`static`** method from **`ClassX`**.

d) A non-**`static`** method from **`ClassX`** may directly reference a **`static`** variable declared inside **`ClassX`**.

e) A **`static`** method from **`ClassX`** may directly reference an instance variable declared inside **`ClassX`**.

f) A **`static`** method from **`ClassX`** may call a non-**`static`** method from **`ClassX`** using the **`this`** keyword for referencing (for example: **`this.method()`**).

g) A **`static`** method from **`ClassX`** may call a non-**`static`** method from **`ClassX`** using the class name for referencing (for example: **`ClassX.method()`**).

Question 3

Give the output of the following syntactically-correct program, assuming the `ExceptionType` classes are subclasses of **`Exception`**.

```
class ExceptionType1 extends Exception{}
class ExceptionType2 extends Exception{}
class ExceptionType3 extends Exception{}
public class ExceptionTest {
      public static void main(String[] args) throws Exception {
            try {
                  A();
                  B();
                  System.out.println("main-try1");
            } catch (ExceptionType1 e) {
                  System.out.println("main-catch1");
            } catch (ExceptionType2 e) {
                  System.out.println("main-catch2");
            } catch (ExceptionType3 e) {
                  System.out.println("main-catch3");
            } finally {
                  System.out.println("mainFinally");
            }
      }
      public static void A() throws Exception {
            try {
                  System.out.println("A1-try");
                  C();
                  System.out.println("A2-try");
            } catch (ExceptionType2 e) {
            } finally {
                  System.out.println("A-finally");
            }
      }
      public static void B() throws Exception {
            System.out.println("B-one");
            throw new ExceptionType2();
      }
      public static void C() throws Exception {
            System.out.println("C-one");
            throw new ExceptionType3();
      }
}
```

```
Output:
Line 1:_____________

Line 2:_____________

Line 3:_____________

Line 4:_____________

Line 5:_____________
```

Question 4

a. Analyze the following code and circle the letter of one choice.

```
public class Test {
  public static void main(String[] args) {
    B b = new B();
    b.m(5);
    System.out.println("i is " + b.i);
  }
}
```

```
class A {
  int i;
  public void m(int i)
{
    this.i = i;
  }
}
class B extends A {
  public void m(String
s) {
  }
}
```

A. The program has a compilation error, because m is overridden with a different signature in B.
B. The program has a compilation error, because b.m(5) cannot be invoked since the method m(int) is hidden in B.
C. The program has a runtime error on b.i, because i is not accessible from b.
D. The method m is not overridden in B. B inherits the method m from A and defines an overloaded method m in B.

b. Which <u>one</u> of the following statements is true?

A. At least one constructor must always be defined explicitly.
B. Every class has a default constructor.
C. Multiple constructors can be defined in a class.
D. A subclass must implement at least one constructor.

c. What is the output of the following code?

```
public class Test {
  public static void main(String[] args) {
    new Human().printHuman ();
    new Student().printHuman ();
  }
}
class Human {
  private String getInfo() {
    return "Human";
  }
  public void printHuman() {
    System.out.println(getInfo());
  }
}
```

```
class Student extends Human {
  private String getInfo() {
    return "Student";
  }
}
```

Question 5

What is printed by the following code?

```
class C1 extends C3 { }
class C2 extends C3 { }
class C3 extends C4 { }
class C4 extends Object { }
public class Test {
     public static void main(String[] args) {
          Object c1 = new C1();
          Object c2 = new C2();
          Object c3 = new C3();
          Object c4 = new C4();
          System.out.println(c1 instanceof C2);//_____________
          System.out.println(c2 instanceof C1);//_____________
          System.out.println(c3 instanceof C1);//_____________
          System.out.println(c3 instanceof C2);//_____________
     }
}
```